

The examination is being carried out on the **following application documents**

Description, Pages

1-47 as published

Sequence listings, SEQ ID NO

1-14 as published

Claims, Numbers

1-15 filed with entry into the regional phase before the EPO

Drawings, Sheets

1/4-4/4 as published

- 1 In respect of the above identified European patent application, only part of the claimed subject-matter has been searched, following an objection of lack of unity of the invention by the European Patent Office acting as International Searching Authority (see Article 17(3)(a) PCT).

In the procedure before the European Patent Office as designated, the Examining Division agrees with the finding of lack of unity of the invention for the following reasons:

The application relates to the use of a substance which stimulates the expression of MC-1R, MC-2R or mu-opioid R, for the preparation of a cosmetic, nutraceutical or pharmaceutical composition and uses thereof for treating e.g ageing of the skin, to increase skin pigmentation or prevent skin depigmentation. The specific substances which are stimulators of the expression of MC-1R, MC-2R or mu-opioid R are claimed in claim 9. In claim 10 the use of these substances for treating the skin, e.g. improve the color of the skin or modify skin pigmentation, is claimed. Independent claim 17 is directed to nucleotides which are sense or antisense sequences of MC - 1R, MC - 2R, mu-opoid receptor or POMC.

The substance or plant extracts which are stimulators of the expression of MC - 1R, MC-2R, mu-opioid receptor (see claim 8) and which are subject-matter of independent claim 10 do not share a significant structural element, nor do they belong to a single class of chemical compounds in the art to which the invention pertains.

The common technical feature is the use of the stimulators of the expression of MC-R1, MC-2R or mu-opioid R for the preparation of cosmetic, nutraceutical or pharmaceutical compositions. This feature is not novel in view of the disclosure of the document US2004/0214851 (see [0020], [0047], claims 34 and 36) which discloses the use of compounds which induce expression of the mu-opioid receptor for reducing scarring of the skin. Furthermore, the prior art discloses the use of *Achillea millefolium* extracts for promoting wound healing (WO2007/030666: page 2 lines 21-30, claim 1) or reducing skin pigmentation (see EP1543825: [0007], claim 4).

In accordance with Rule 164(2) EPC the applicant is invited to limit the application to the invention covered by the international search report.

The subject-matter to be excised may be made the subject of one or more divisional applications according to Rule 36 EPC.

2 Art. 84 EPC:

- 2.1 Claim 1 relates to substance which are stimulators of the expression of MC-R1, MC-R2 or mu-opioid R. The functional definition of the substances does not allow to ascertain the scope of the claim. Moreover, the definition encompasses a large number of compounds and the disclosure is not sufficiently precise for the person skilled in the art to reduce the technical features to practice without undue burden. Claim 1 does not meet the requirement of Article 84 EPC.
- 2.2 Claim 1 is directed to the use of a compound as an active ingredient in a cosmetic composition or for the preparation of nutraceutical composition. Claims 4-5 are second medical use claims drafted in the swiss-type claim. Dependent claims 6-11 are dependent of both claim 1 and claims 4-5 thus covering both therapeutic and cosmetic uses. The scope of claims 6-11 is thus unclear.
- 2.3 It is not clear which cosmetic use or therapeutic indications fall in the scope of the definition of claims 7, 10, 12 and 13. These claims encompass both therapeutic and cosmetic uses. The scope of said claims is unclear.

2.4 It is not clear which compounds fall in the functionally defined compounds in of claim 11. The scope of claim 11 is unclear.

3 Prior art:

Reference is made to the following documents; the numbering will be adhered to in the rest of the procedure.

D1 US 2004/214851 A1 (BIRMACHU WOUBALEM M R [US] ET AL) 28 October 2004 (2004-10-28)

D2 EP 1 543 825 A (KURARAY CO [JP]; POLA CHEM IND INC [JP]) 22 June 2005 (2005-06-22)

D3 WO 2007/030666 A (HERBAL INTERVENTION CORP [US]; MULVANERTY NOREEN [US]) 15 March 2007 (2007-03-15)

D4 US 3 522 350 A (GOLDBERG ARTHUR ET AL) 28 July 1970 (1970-07-28)

D5 DE 201 12 636 U1 (KUNDIC SIDA [DE]) 10 January 2002 (2002-01-10)

D6 FR 2 578 422 A (CARIEL LEON [FR]) 12 September 1986 (1986-09-12)

If not indicated otherwise the relevant passages are those mentioned in the search report.

Document D1 discloses the use of compounds inducing the expression of mu - opioid receptor for making pharmaceutical compositions for treating lesions of the skin.

Document D2 discloses the use of compounds extracted from *Achillea millefolium* for lightening of the skin.

Document D3 discloses the use of plant extracts inter alia extracts of yarrow, eventually in combination with other compounds like growth factors, for promoting healing of wounds of the skin.

Document D4 discloses compositions of yarrow extract have anti-inflammatory activity when applied topically to the skin.

Document D5 discloses compositions for treatment of the skin in particular wounds, comprising an extract of *Achillea millefolium*.

Document D6 discloses extracts of *Achillea millefolium* for treating skin hyper pigmentation.

4 Novelty (Art. 54 EPC):

4.1 It is noted that the presence of a mechanism of action i.e. stimulation of the expression of MC-R1, MC-R2 or mu-opioid receptors does not delimit a claim from the prior art. Therefore, it is considered that claims 1, 4 and 5 lack novelty over D1-D6.

4.2 Claim 8 lacks novelty in view of D2-D6.

4.3 Claims 12-14 are not novel in view of D2-D6.

5 Inventive step (Art. 56 EPC):

Dependent claims 2-3, and 11-13 do not contain any features which, in combination with the features of any claim to which they refer, meet the requirements of the EPC in respect of novelty and/or inventive step, the reasons being as follows: the subject-matter of said claims is either disclosed or rendered obvious by documents D1-D6. With regard to the combinations of claims 11-13 it is noted that the application does not provide any evidence of an unexpected technical effect due to the combination. It would be obvious for a skilled man to combine a stimulator of MC-R1, MC-R2 or mu-opioid receptor expression e.g. an extract of *Achillea millefolium* with a second compound as claimed.

6 The third-party observation received on 30.06.2010 pursuant to Art. 115 EPC; in particular exhibit 1 calls into question the patentability of the subject-matter claimed for the reasons given below. Thus, exhibit 1 will be taken into account in the proceedings (Guidelines E-VI, 3) and the numbering will be adhered to in the rest of the procedure.

Exhibit 1 discloses a composition for the treatment of skin marks comprising *Achillea millefolium* and appears thus to be relevant for the novelty and/or inventive step of present claims.

7 The applicant is invited to file new claims which take account of the above comments.

In order to facilitate the examination of the conformity of the amended application with the requirements of Article 123(2) EPC, the applicant should clearly identify the amendments carried out, irrespective of whether they concern amendments by addition, replacement or deletion, and to indicate the passages of the application as filed on which these amendments are based.